Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when assumptions apply.
3. Explain energy and model limits clearly.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to explain your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

A model target needs a 3 km sideways shift over 5×10^{6} s. Its mass is 7×10^{8} kg.
Estimate Δv and impulse; state the model used.

Word help: estimate = find an approximate value

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

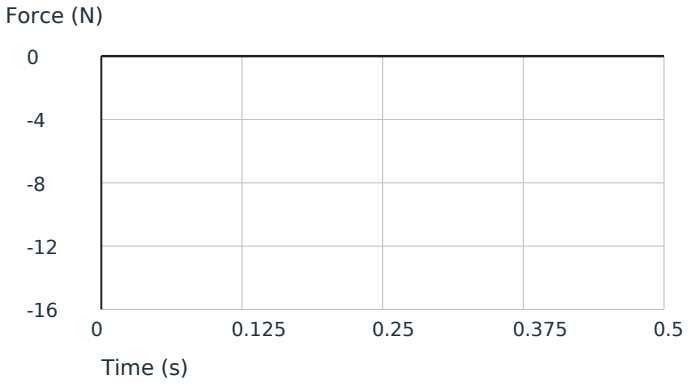
02 Worked example: explain the model

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. Label before/after diagrams and directions.

Worked reasoning: A 2 kg cart moving at 3 m/s stops in 0.5 s. Momentum change is -6 kg·m/s; average force is -12 N. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A 1 kg cart at 4 m/s right sticks to a 1 kg cart at rest. **Draw** before/after and explain which quantity is conserved.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Under the assumption …, the evidence … supports … because … .

Before contact: add velocity arrows.



Marzano check - circle: 1 2 3

04 Apply the physics

A 2 kg cart at +6 m/s sticks to a 1 kg cart at -3 m/s. **Find** final velocity and kinetic energy transformed. Organize the known quantities on the before-and-after model.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A report uses equal total momentum before/after to claim a collision is elastic. What additional calculation is required?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart has momentum -24 kg·m/s and mass 4 kg. **Find** velocity and its new momentum if velocity reverses at unchanged speed. **Use** before and after snapshots to organize the momentum account.

Word help: velocity = rate of position change, with direction

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

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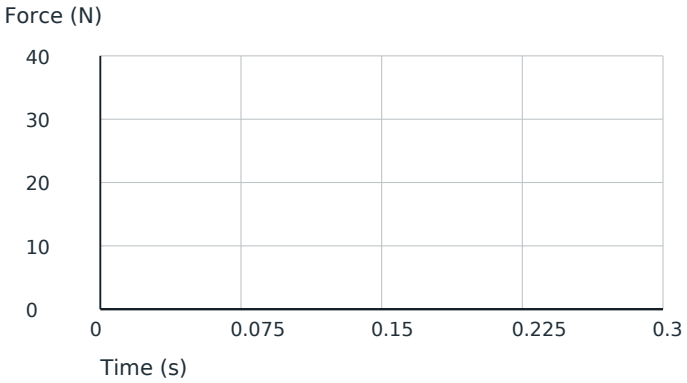
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

A triangular force pulse has peak 40 N and duration 0.30 s. It acts on a 2 kg cart initially at -1 m/s.
Find impulse and final velocity. Organize the known quantities on the before-and-after model.

Sentence starter: I found ... [unit]. This value means ... in this situation.



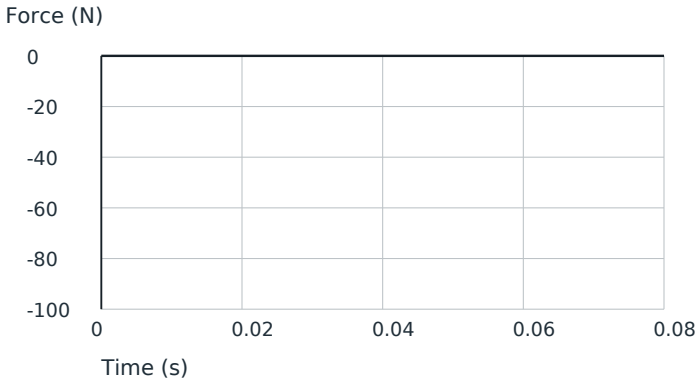
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.4 kg ball rebounds from +5 to -3 m/s. A triangular force pulse lasts 0.08 s.
Find impulse, average force and peak-force magnitude. Organize the known quantities on the before-and-after model.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3